

Trend-Following Strategy Suite — Backtest Report

Moving-average crossover, Donchian channels, and the Turtle system · real market data · generated 2026-08-02

Real market data: this run uses real daily prices for AMD, NFLX, NVDA, PLTR, TSLA (2025-08-01 to 2026-07-31, 251 trading days), loaded from local CSV files. That's 5 stocks over about one year — a small sample, not the full SP100 or a multi-year history. Treat results as a pipeline demonstration and a first look, not a final strategy recommendation.

Executive Summary

Tickers tested: **AMD, NFLX, NVDA, PLTR, TSLA** · Period: **2025-08-01 to 2026-07-31** (251 trading days) · Portfolio construction: **inverse-volatility weighted**, scaled to a **10% annualized volatility** target per strategy variant.

Best-ranked strategy (by Sharpe ratio): **MAC -- Slow (32/128 day EWMA)** — annualized return +2.7%, Sharpe ratio 0.27, max drawdown -7.3%.

Portfolio-Level Comparison (ranked best to worst)

Rank	Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
1	MAC -- Slow (32/128 day EWMA)	+2.7%	+10.0%	0.27	-7.3%	0.37	0.35
2	Turtle -- System 2 (55-day entry / 20-day exit)	-1.0%	+10.0%	-0.10	-11.5%	-0.09	-0.14
3	Turtle -- System 1 (20-day entry / 10-day exit)	-3.2%	+10.0%	-0.32	-8.7%	-0.37	-0.44
4	MAC -- Medium-Fast (4/16 day EWMA)	-5.9%	+10.0%	-0.59	-7.5%	-0.79	-0.78

Rank	Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
5	MAC -- Fast (2/8 day EWMA)	-6.0%	+10.0%	-0.60	-9.0%	-0.67	-0.83
6	Equal Blend (all 9 variants)	-4.9%	+7.7%	-0.64	-7.5%	-0.65	-0.83
7	MAC -- Medium-Slow (16/64 day EWMA)	-7.1%	+10.0%	-0.71	-9.5%	-0.74	-0.93
8	Donchian -- 20-day channel	-7.2%	+10.0%	-0.72	-15.8%	-0.45	-0.95
9	MAC -- Medium (8/32 day EWMA)	-8.2%	+10.0%	-0.82	-10.5%	-0.78	-1.07
10	Donchian -- 55-day channel	-9.3%	+10.0%	-0.93	-12.3%	-0.76	-1.19

Per-Stock Detail (each strategy variant, each stock)

AMD

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Fast (2/8 day EWMA)	+32.0%	+71.8%	0.45	-42.6%	0.75	0.71
MAC -- Medium-Fast (4/16 day EWMA)	+8.9%	+71.8%	0.12	-50.9%	0.18	0.19
MAC -- Medium (8/32 day EWMA)	+10.0%	+71.8%	0.14	-52.3%	0.19	0.22
MAC -- Medium-Slow (16/64 day EWMA)	-11.5%	+71.8%	-0.16	-56.9%	-0.20	-0.23
MAC -- Slow (32/128 day EWMA)	-22.8%	+71.8%	-0.32	-62.4%	-0.37	-0.46
Donchian -- 20-day channel	+25.8%	+70.2%	0.37	-58.0%	0.45	0.57
Donchian -- 55-day channel	-15.2%	+62.6%	-0.24	-53.1%	-0.29	-0.36
Turtle -- System 1 (20-day entry / 10-day exit)	+43.2%	+48.0%	0.90	-24.3%	1.78	1.54

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
Turtle -- System 2 (55-day entry / 20-day exit)	+39.9%	+44.4%	0.90	-21.7%	1.83	1.53

NFLX

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Fast (2/8 day EWMA)	+3.7%	+34.9%	0.11	-36.4%	0.10	0.15
MAC -- Medium-Fast (4/16 day EWMA)	-1.5%	+34.9%	-0.04	-38.5%	-0.04	-0.06
MAC -- Medium (8/32 day EWMA)	+3.8%	+34.9%	0.11	-28.7%	0.13	0.14
MAC -- Medium-Slow (16/64 day EWMA)	+0.5%	+34.9%	0.01	-28.5%	0.02	0.02
MAC -- Slow (32/128 day EWMA)	+29.9%	+34.8%	0.86	-32.0%	0.93	1.20
Donchian -- 20-day channel	+24.9%	+34.2%	0.73	-28.9%	0.86	1.02
Donchian -- 55-day channel	+3.0%	+31.6%	0.10	-36.7%	0.08	0.13
Turtle -- System 1 (20-day entry / 10-day exit)	+20.9%	+22.7%	0.92	-19.8%	1.06	1.38
Turtle -- System 2 (55-day entry / 20-day exit)	+9.2%	+26.7%	0.35	-29.1%	0.32	0.45

NVDA

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Fast (2/8 day EWMA)	-53.9%	+36.1%	-1.49	-53.8%	-1.00	-1.87
MAC -- Medium-Fast (4/16 day EWMA)	-63.2%	+35.9%	-1.76	-63.1%	-1.00	-2.14
MAC -- Medium (8/32 day EWMA)	-50.6%	+36.2%	-1.40	-50.5%	-1.00	-1.79

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Medium-Slow (16/64 day EWMA)	-58.7%	+36.0%	-1.63	-58.6%	-1.00	-2.04
MAC -- Slow (32/128 day EWMA)	-19.3%	+36.4%	-0.53	-31.1%	-0.62	-0.74
Donchian -- 20-day channel	-48.1%	+35.4%	-1.36	-50.5%	-0.95	-1.77
Donchian -- 55-day channel	-29.7%	+32.9%	-0.90	-35.3%	-0.84	-1.24
Turtle -- System 1 (20-day entry / 10-day exit)	-27.8%	+12.6%	-2.20	-27.7%	-1.00	-2.54
Turtle -- System 2 (55-day entry / 20-day exit)	-13.0%	+8.7%	-1.49	-13.6%	-0.96	-1.75

PLTR

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Fast (2/8 day EWMA)	-14.2%	+52.4%	-0.27	-40.9%	-0.35	-0.38
MAC -- Medium-Fast (4/16 day EWMA)	-39.0%	+52.4%	-0.75	-48.6%	-0.80	-1.02
MAC -- Medium (8/32 day EWMA)	-57.9%	+52.2%	-1.11	-64.5%	-0.90	-1.48
MAC -- Medium-Slow (16/64 day EWMA)	-33.2%	+52.4%	-0.63	-51.6%	-0.64	-0.88
MAC -- Slow (32/128 day EWMA)	-4.9%	+52.4%	-0.09	-38.0%	-0.13	-0.13
Donchian -- 20-day channel	-67.9%	+48.9%	-1.39	-74.6%	-0.91	-1.80
Donchian -- 55-day channel	-25.3%	+47.3%	-0.53	-42.2%	-0.60	-0.75
Turtle -- System 1 (20-day entry / 10-day exit)	-30.0%	+24.6%	-1.22	-31.3%	-0.96	-1.64
Turtle -- System 2 (55-day entry / 20-day exit)	-16.2%	+13.9%	-1.17	-17.2%	-0.94	-1.40

TSLA

Strategy / Variant	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Calmar	Sortino
MAC -- Fast (2/8 day EWMA)	-31.3%	+46.3%	-0.68	-53.2%	-0.59	-0.95
MAC -- Medium-Fast (4/16 day EWMA)	+48.5%	+46.2%	1.05	-33.8%	1.43	1.58
MAC -- Medium (8/32 day EWMA)	-3.6%	+46.3%	-0.08	-47.5%	-0.08	-0.11
MAC -- Medium-Slow (16/64 day EWMA)	+16.9%	+46.3%	0.36	-45.0%	0.37	0.54
MAC -- Slow (32/128 day EWMA)	-1.3%	+46.3%	-0.03	-48.5%	-0.03	-0.04
Donchian -- 20-day channel	+6.7%	+44.5%	0.15	-36.0%	0.19	0.22
Donchian -- 55-day channel	-47.2%	+36.4%	-1.30	-49.9%	-0.95	-1.61
Turtle -- System 1 (20-day entry / 10-day exit)	-11.2%	+26.1%	-0.43	-23.0%	-0.49	-0.57
Turtle -- System 2 (55-day entry / 20-day exit)	-15.1%	+12.5%	-1.21	-16.2%	-0.93	-1.40

Methodology & Assumptions

- **Three strategy families, nine variants total.** *MAC* (dual exponentially-weighted-moving-average crossover, five speeds, Man AHL-style): long when the fast average is above the slow average, short otherwise. *Donchian channel* (two window lengths, 20 and 55 days): long on a breakout above the highest high of the prior window, short on a breakout below the lowest low, holding until the opposite breakout. *Turtle* (Richard Dennis's System 1 and System 2): the same breakout entries as Donchian, plus volatility-scaled position sizing (via Average True Range), pyramiding up to 4 units as a winning trade extends, and a stop-loss at 2x ATR.
- **Parameter choices:** MAC's five fast/slow day-pairs (2/8, 4/16, 8/32, 16/64, 32/128) are a standard doubling-ladder stand-in, not Man AHL's own undisclosed exact parameters. Donchian and Turtle use 20/55-day windows, the periods most associated with the original Turtle system.
- **Portfolio weighting:** inverse-volatility (each stock's weight is proportional to 1/its own volatility, so every stock contributes roughly equal risk, not equal share count), then the whole portfolio is scaled to 10% annualized volatility per variant — so all nine variants are compared on equal footing.
- **"Equal Blend"** now averages all nine variants across all three strategy families equally weighted, not five same-family speeds as in an earlier MAC-only version of this report — a cruder blend than averaging within one strategy family, included for reference rather than as a recommended combination.
- **Risk-free rate:** Sharpe and related ratios assume a 0% risk-free rate, not a real T-bill rate.

- **No look-ahead bias:** each day's trading signal (MAC's crossover state, a Donchian/Turtle breakout level or ATR) is decided using only prices known by that day's close, and only affects the following day's return.
- **Turtle simplifications:** no "skip this breakout if the last trade in this direction was a winner" filter (a real but often omitted refinement); position sizing is a fraction of one strategy-level unit, not real share counts or dollar risk; no cross-instrument correlation or portfolio-heat cap (handled instead, approximately, by this report's own inverse-volatility portfolio weighting across strategies).
- **Sample size:** only 5 tickers and about one year of real data tested here (a first real-data run, not yet the full SP100 or a multi-year history); the same pipeline scales to more tickers and a longer window once more data is loaded.

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